

Please group together with your Operating System

MacOS?

-> Please sit to the left

Linux?

-> Please sit in the middle

Windows?

-> Please sit to the right

With funding from the:

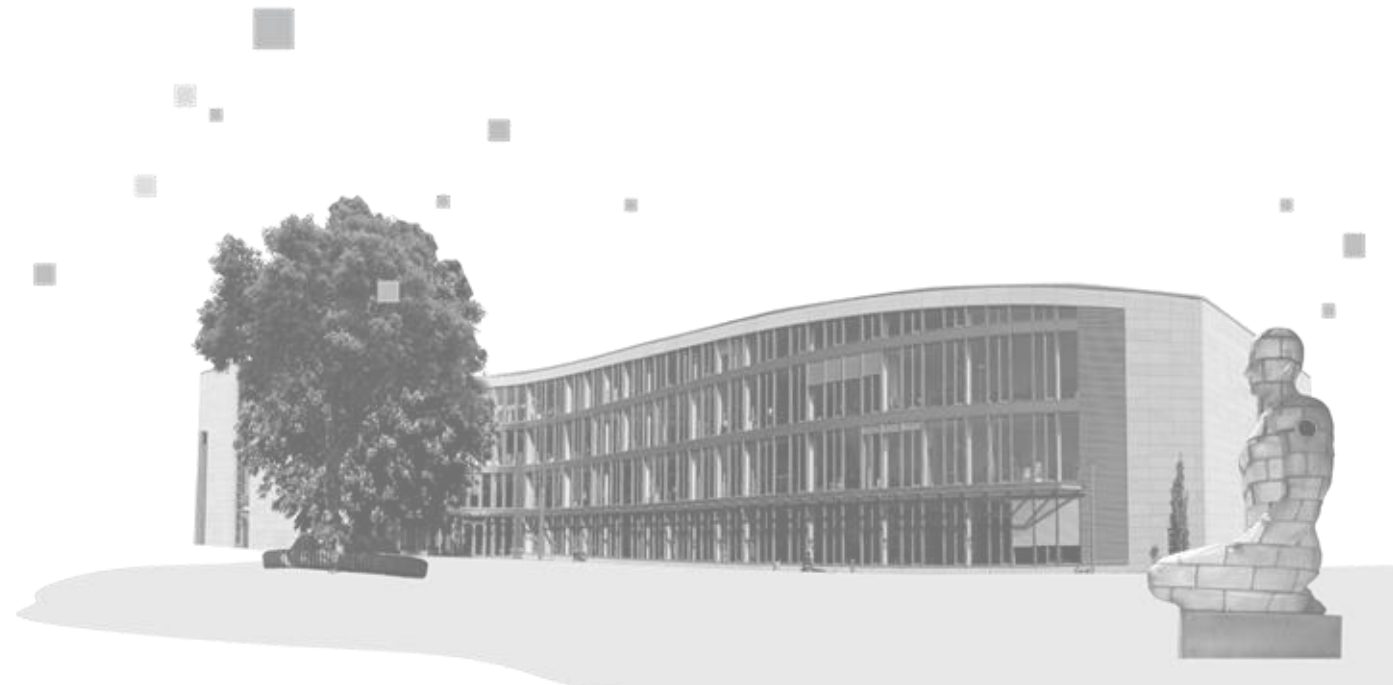


First Steps Towards Your Own AI Application

Setting up the key tools for local AI development

**Design IT.
Create Knowledge.**

www.hpi.de



Overview of the AI Service Center

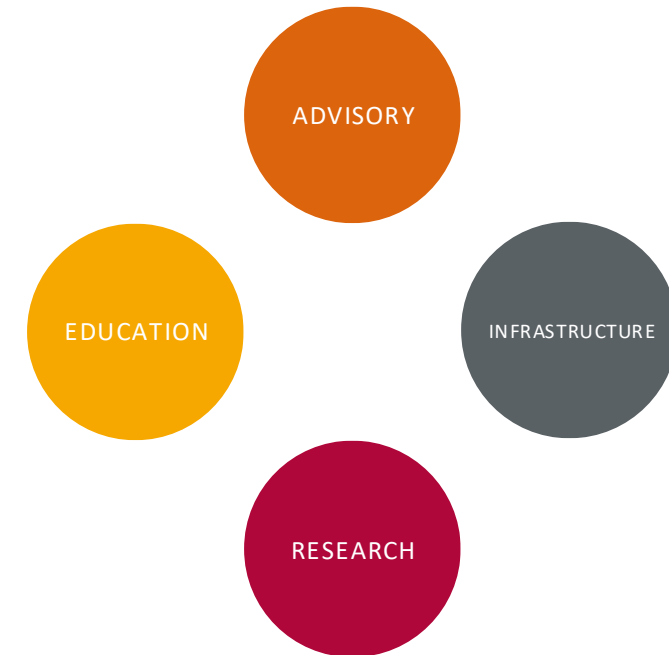
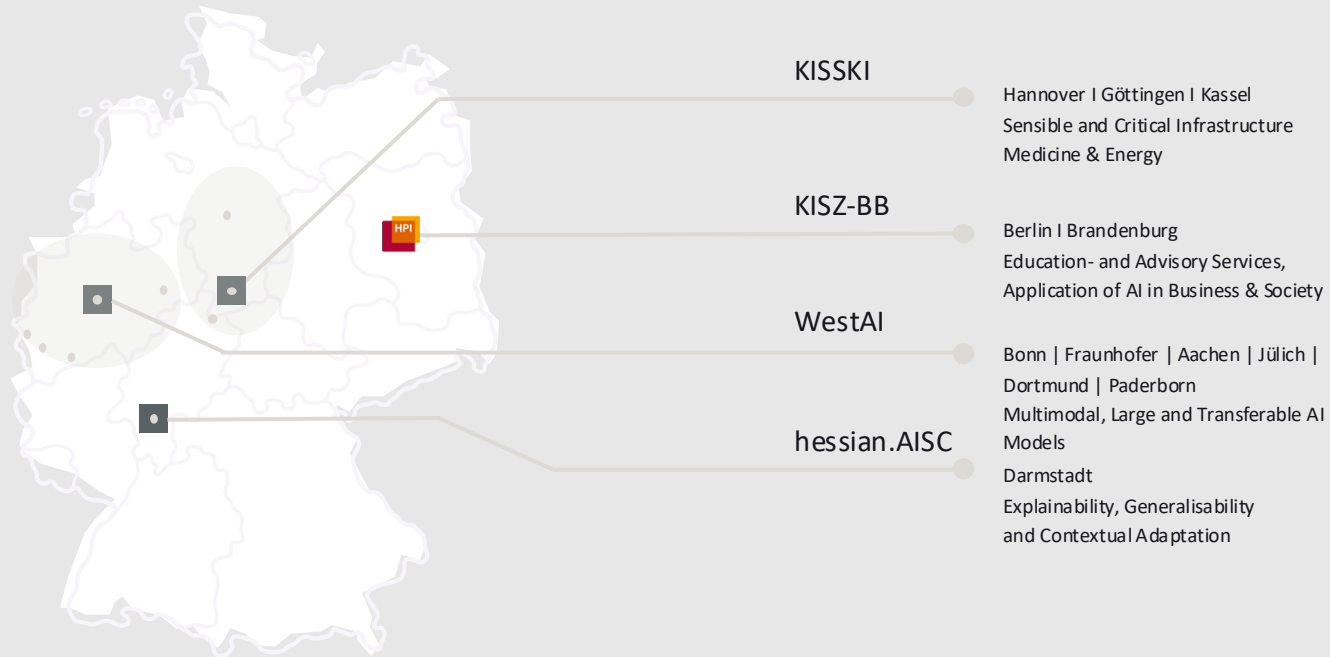


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Federal Ministry
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and Space

AI-Service Centers



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ADVISORY

AI Advisory

- General Questions regarding AI,
- Computing Infrastructure,
- Model, Architecture, Framework Selection

AI Pilot Projects (3 months)

1. Understanding and Identifying the Use Case
2. Specifying the Use Case
3. Developing a Solution
4. Finalizing and Documenting

Cooperations

- Co-organized Network Meetings or Workshops

Start-up Support

- All of the above
- Support with Networking

HPI ECOSYSTEM



d—school

Engine 



Mittelstand-Digital
Zentrum
Berlin

Infrastructure



Training

- 8 x 8 NVIDIA H100 GPU

Inference

- 5 x 8 NVIDIA A30 GPU

ARM Server

- Ampere Altra Max M128-30 CPU
- 2 x NVIDIA L40 GPUs

GPU Server

- AMD Epyc CPU
- 8 x NVIDIA L40S GPU

Edge

- ARMv8 CPU
- NVIDIA Jetson AGX Module

Storage

- 1.5 PB NVRAM

Network

- 400 Gb/s Infiniband
- 200 Gb/s Ethernet

kisz@hpi.de

hpi.de/kisz

EDUCATION & TRAINING

Talks



Guest Talk: Automatische Deepfake-Erkennung mit KI

Di. 17. März, 17:00

Hasso-Plattner-Institut

Kostenlos

MOOCs



Chatbots leicht gemacht: Große Sprachmodelle einfach nutzen

📅 SELF-PACED SINCE DEC 10, 2025

DATA SCIENCE

🏆 RECORD OF ACHIEVEMENT

🌐 DE

Workshops

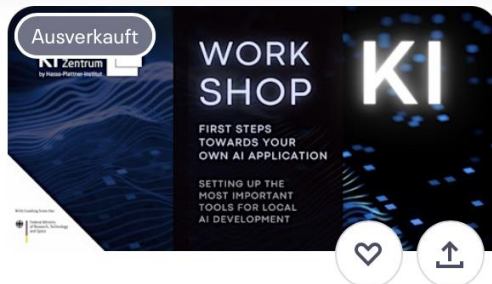


Reinforcement Learning – Part II

Do. 26. Feb., 10:00

Hasso-Plattner-Institut

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First Steps Towards Your Own AI Application

Donnerstag um 15:00

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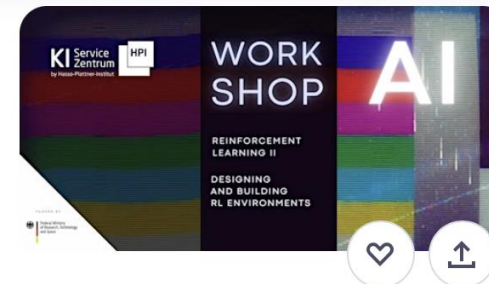


Effizientes Abfragen von Informationen – Teil II

Mi. 25. Feb., 13:00

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Reinforcement Learning – Part II

Do. 26. Feb., 10:00

Hasso-Plattner-Institut

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Erste Schritte zu Ihrer eigenen KI-Anwendung

Fr. 13. März, 13:00

Hasso -Plattner-Institut

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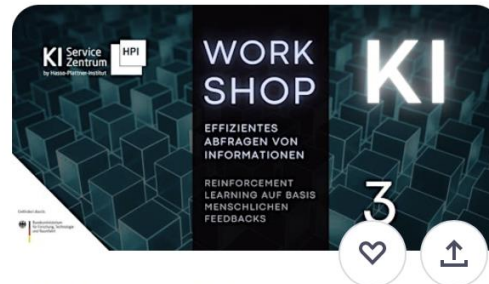


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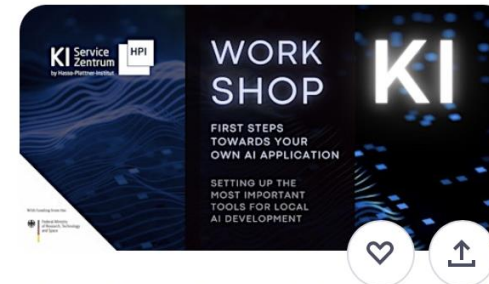


Effizientes Abfragen von Informationen – Teil III

Do. 26. März, 14:00

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First Steps Towards Your Own AI Application

Mo. 13. Apr., 10:00

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Effizientes Abfragen von Informationen – Teil IV

Di. 28. Apr., 15:00

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Let's get to know each other

- What is your background?
- What is your motivation coming here today?
- Are you familiar with VSCode, Git, Docker, Python, ... ?

With funding from the:

Goal: Set up your environment

1. VS Code
2. WSL
3. Git
4. Python
5. UV
6. Ollama
7. Docker Desktop
8. Example Application: Local Chatbot

With funding from the:

Overview of the software

AISC Getting Started - AI Chatbot Demo

Chat with a local AI model powered by Ollama

AI Chat Interface

● Connected to Backend   Clear



Hello! I'm an AI assistant powered by Ollama running locally. Ask me anything!

which model are you?



I'm Llama 3.2 1B. It's a large language model developed by Meta, designed to provide helpful and informative responses to your questions.

Are you running locally on my laptop?



Yes, I'm running locally on your device. This means that the AI was trained on a dataset of text from the internet, which is then used to generate human-like responses to your queries. Your language input is then passed through my network of algorithms and models, which produce the output you see.

Type your message here... (Press Enter to send)



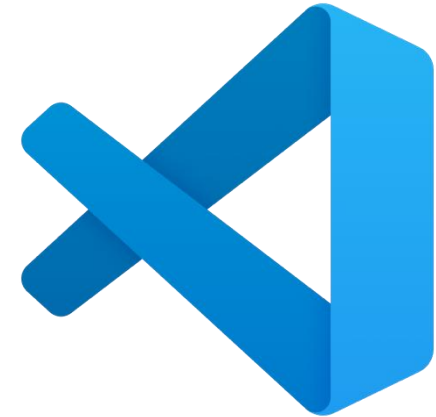
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Visual Studio Code (VS Code)

- Code-Editor:
 - Lightweight
 - Cross-platform
 - Expandable with extensions
- Supports various tools:
 - Python, Jupyter Notebook, Git, Docker, SSH
- Integration of coding assistants (e.g., Claude)



GitHub & Git

- Platform for version control of code
 - Development, version control, review, and documentation



With funding from the:



Python

- General-purpose programming language with key libraries for AI and data science
 - NumPy: numerical computing
 - PyTorch: neural networks
 - Librosa: audio processing
 - ...
- Extensive learning resources and tutorials available
- Suitable for data processing, model training, ...



UV

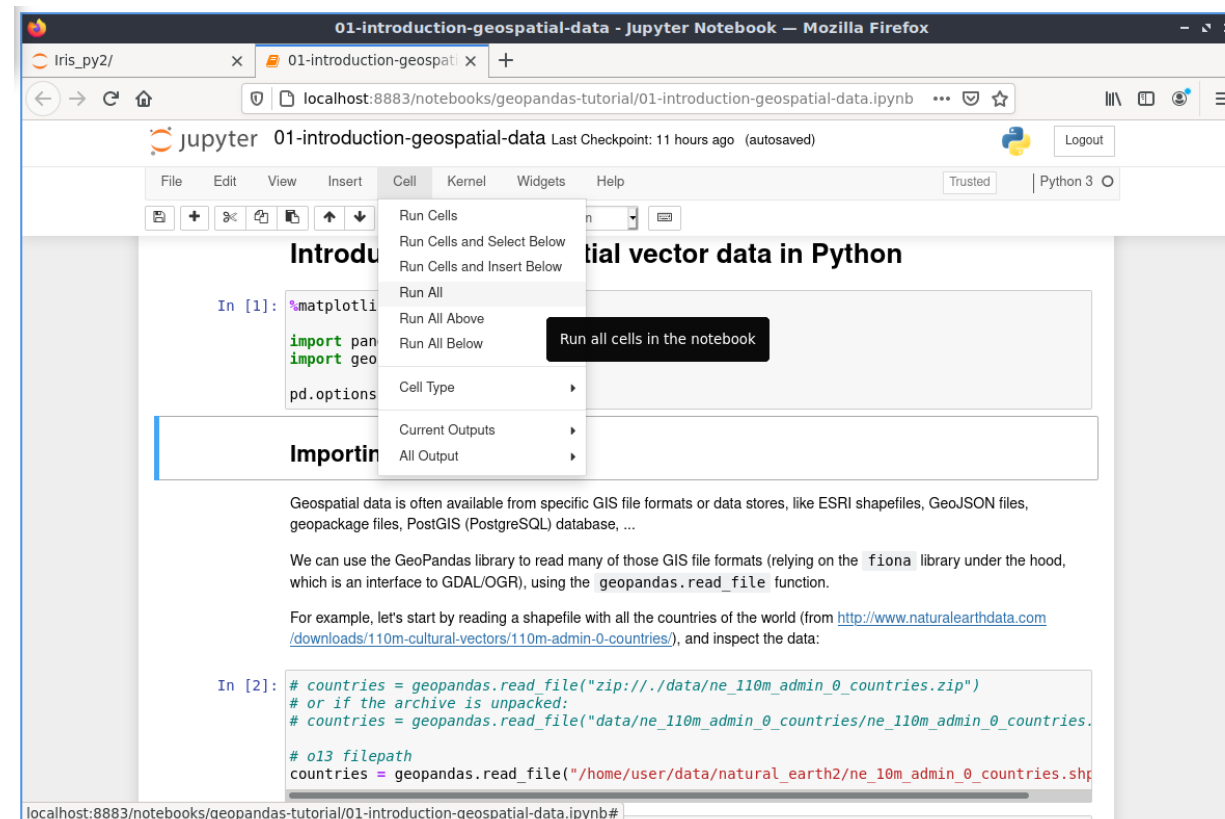
- Fast Python package and project manager
 - Handles package installation and dependency management
 - Creates reproducible development environments, ensuring consistent package versions
- Ensures code runs reliably across different machines and operating systems
 - Windows 10, 11, XP
 - Linux Mint, Ubuntu, Debian
 - MacOS



With funding from the:

Jupyter notebooks

- Supports code, narrative text, equations (Markdown), and visualizations
- Ideal for exploratory data analysis, prototyping, and teaching
- Code cells can be executed independently --> Easy debugging



Docker

- Container platform for packaging reproducible runtime environments
 - Enables applications to run consistently across machines, servers, and high-performance clusters (HPC)
 - Solves "works on my machine" problem
- Ideal for complex software stacks



With funding from the:

<https://github.com/aihpi/workshop-getting-started>

Gefördert durch:



AI for Everybody — Getting Started

This repository hosts the **Getting Started** workshop offered by the [AI Service Centre Berlin Brandenburg](#). It introduces the development tools and workflows — VSCode, Git, WSL, Python, UV, Ollama, and Docker — that participants will use across our workshops.

Workshop materials

All workshop content lives in [03_workshop/](#). The workshop's own README is the entry point:

[03_workshop/README.md](#)

It contains the agenda, the per-tool guides, the Jupyter notebooks for replay at home, and the slides.

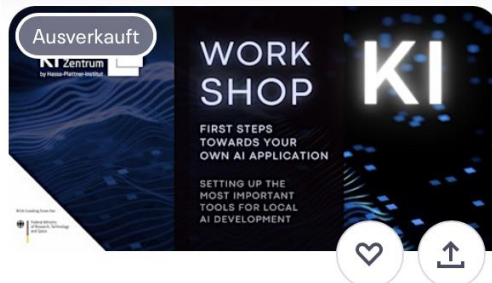
Example application

This repository also includes a small chatbot application that demonstrates what you can build once your environment is set up:

- [01_frontend/](#) — React chat interface
- [02_backend/](#) — FastAPI server with Ollama integration
- [docker-compose.yml](#) — runs the full stack locally

The chatbot is the final demo of the workshop (section 8). See [03_workshop/guides/08_chatbot_demo.md](#) for the walkthrough.

Finished? Check out our upcoming workshops, talks and online courses!



First Steps Towards Your Own AI Application

Donnerstag um 15:00

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Effizientes Abfragen von Informationen - Teil II

Mi. 25. Feb., 13:00

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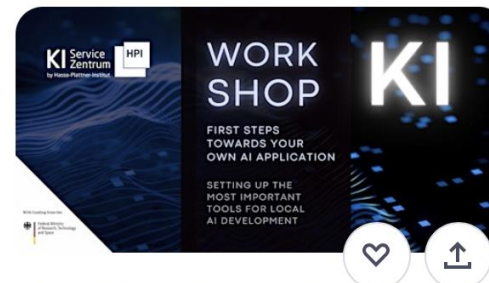


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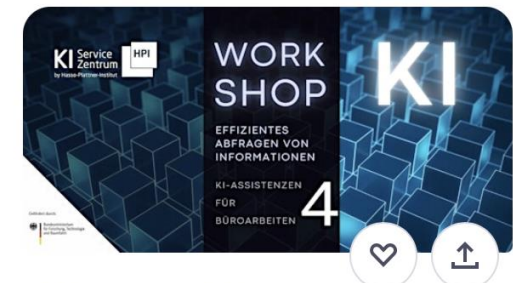


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Effizientes Abfragen von Informationen - Teil IV

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Finished? Check out our upcoming workshops, talks and online courses!

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Digitale Gerechtigkeit - Wie du dich für eine fairere KI-Zukunft engagieren...

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Efficient AI for Weather Forecasting

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Chatbots leicht gemacht: Große Sprachmodelle einfach nutzen

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Lehrkräfteprofessionalisierung für KI in Schule und Unterricht

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PROFESSIONAL SKILLS

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Argumentieren lernen mit KI - für Schule & Alltag

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Time Series Analysis

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Sustainability in the Digital Age: Efficient AI Techniques in the LLM Era

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Practical Computer Vision with PyTorch

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KI-Biases verstehen und vermeiden

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Profitable AI

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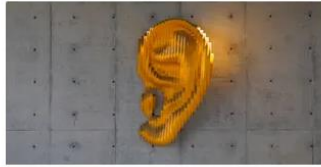
Understanding Embeddings for Natural Language Processing

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CONFIRMATION OF PARTICIPATION

EN



Selection and Implementation of AI Models: Speech2Summary

SELF-PACED SINCE SEP 30, 2023

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important!



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form



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